Pollution to Baking Soda Scrubber Project

Comprehensive 2-Page Executive Summary: Engineering Specifications, Chemical Kinetics & Financial Model

1. PROJECT OVERVIEW & CORE MECHANISM

The **Pollution to Baking Soda Scrubber** is an advanced industrial air pollution control and eco-friendly byproduct valorization system. Designed to capture hazardous flue gases and convert them into commercially viable high-purity baking soda ($NaHCO_3$) and neutralized mineral-rich effluents, the system achieves active ecosystem protection while maintaining complete financial self-sustainability without relying on external tipping fees.

System Architecture & Parts

- **Intake Duct (0.8m dia):** Transfers flue gas at turbulent $v = 5.5\ m/s$.
- **Dust Filter (25 m²):** Removes heavy particulate matter.
- **Acid Filter (45 m²):** Carbon mesh capturing SO_X & NO_X while letting CO_2 pass.
- **Scrubber Tower:** 25m length × 3.5m diameter chamber with high-pressure atomizers.
- **Heater Floor (8 m²):** Precision thermal drying ($40°C–45°C$).

Operational Workflow

1. **Flue Gas Intake:** Industrial exhaust (10,000 m³/h) drawn through intake pipe.
2. **Filtration:** Dust and acid filters strip particulate matter and sulfur/nitrogen oxides.
3. **Carbon Separation:** Pure CO_2 enters the 25m reaction tower.
4. **Precipitation Spray:** Continuous rain of Na_2CO_3 solution ($15\ \mu m$ droplets) reacts to form wet baking soda slurry.
5. **Thermal Drying:** Heated floor removes 15–18% moisture without thermal decomposition.

2. INDUSTRIAL SCALE-UP & ENGINEERING PARAMETERS

Parameter / Specification	Design Value	Engineering Function & Impact
Industrial Gas Flow Rate (Q)	10,000 m³/h	Handles heavy industrial flue gas output continuously.
Scrubber Tower Volume	240.5 m³ (3.5m dia × 25m len)	Provides 86.5 seconds of optimal gas-liquid residence time.
Spray Solution Flow & Droplets	250 L/h ($15\ \mu m$ mist)	Generates 100,000 m² of active liquid mist surface area.
Chemical Consumption	106 kg/h Na_2CO_3	Reacts with ~100 kg/h CO_2 input.
Raw Baking Soda Output	147.8 kg/h slurry	Yields 1,064.16 metric tons annually over 7,200 operating hours.
Heater Floor Power & Temp	16.5 kW ($40°C–45°C$)	Evaporates 25 kg/h moisture; strictly capped below $50°C$ to prevent decomposition.

3. ENERGY BALANCE & POWER SYSTEMS

To ensure robust operation, the plant integrates a high-efficiency electrical and solar architecture:

- **Electrical Load:** High-pressure atomizing pump requires a 1.5 kW motor; total continuous electrical system load reaches 18–20 kW.
- **Solar Array & Storage:** Scaled via a 45 kW photovoltaic array coupled with a heavy-duty industrial 1,000 Ah at 48 V battery bank for uninterrupted backup.

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Part 2: Commercial Economics, CAPEX, OPEX & Profitability Analysis (v10 Model)

4. COMPREHENSIVE CAPITAL EXPENDITURE (CAPEX) BREAKDOWN

Turnkey plant setup utilizing imported marine-grade alloy and corrosion-resistant Hastelloy/PVDF linings:

Equipment / Component	Specifications & Sourcing Origin	Cost in INR (Inclusive of Duties & Freight)
Primary Scrubber Tower & Ducting	3.5m dia × 25m height (European Marine-Grade Alloy / Hastelloy)	₹ 65,00,000
Heavy-Duty Acid Scrubber Unit	PVDF-lined internal walls, Hastelloy spray nozzles & FRP shell	₹ 6,00,000
High-Pressure Pump System	Japanese Industrial Multi-Stage Pumps & Atomization Nozzles	₹ 14,00,000
Baghouse Dust Filter & Carbon Mesh	German High-Efficiency Ceramic Filters & Carbon Absorption Cartridges	₹ 18,50,000
Heater Floor & Drying System	American Thermal Heating Elements & Precision Control Floor	₹ 21,00,000
Solar PV Array & Battery Backup	Tier-1 Imported Monocrystalline PV Panels & Hybrid Inverters	₹ 28,00,000
Automation, Piping & Supervision	Swiss PLC Hardware, Precision Sensors & Expert Supervision	₹ 23,50,000
Total Initial CAPEX	**Turnkey Plant Setup + Heavy-Duty Acid Scrubber Unit**	**₹ 1,76,00,000 (~₹ 1.76 Crore)**

5. OPERATING EXPENDITURE (OPEX) & 3-SHIFT PAYROLL

Calculated for 7,200 operating hours annually (24/7 across 3 shifts daily, 300 active days):

Expense Item	Consumption Rate & Shift Detail	Hourly Cost (₹)	Annual Cost (7,200 hrs) (₹)
Raw Material (Soda Ash / Na_2CO_3)	106 kg/h @ ₹30/kg (Imported high-purity grade)	₹ 3,180	₹ 2,28,96,000
Electrical & Thermal Power	20 kW total load @ ₹7.5/kWh	₹ 150	₹ 10,80,000
Maintenance & Acid Seals	Scheduled replacement of corrosion-resistant seals & filters	₹ 1,150	₹ 82,80,000
24/7 Shift Manpower	3 shifts × [2 Operators + 1 Engineer + 1 Manager]	₹ 600	₹ 43,20,000
Total Annual OPEX	**Continuous Operation & 3-Shift Rotation**	**₹ 5,080 / hr**	**₹ 3,65,76,000 / year**

6. REVENUE PROJECTIONS & FINANCIAL RETURN SUMMARY

Revenue Streams

- **Primary Product (Baking Soda):** 147.8 kg/h yields 1,064.16 metric tons/year @ ₹30/kg = **₹ 3,19,24,800**.
- **Secondary Byproduct (Effluent):** Neutralized SO_X/NO_X solution (500 L/h) @ ₹3/L = **₹ 1,08,00,000**.
- **External Tipping Fees:** Completely removed (₹0).
- **Total Gross Revenue: ₹ 4,27,24,800 (~₹ 4.27 Crore)**.

Financial Summary

Total Gross Revenue	₹ 4,27,24,800 (100.0%)
Total Annual OPEX	-₹ 3,65,76,000 (85.6%)
Net Operating Profit	**₹ 61,48,800 (~₹ 61.5 Lakhs)**
Net Profit Margin / Payback	**14.4% Margin | ~2.8 Years Payback**

Conclusion: Self-sustaining commercial production delivers robust returns with zero dependence on tipping subsidies.